Conditional Distributions

Conditional Distributions — Discrete

- Let X_1 and X_2 have the joint pmf $p_{X_1,X_2}(x_1, x_2)$ with support $\mathcal{S}$. $p_{X_1}(x_1)$ and $p_{X_2}(x_2)$ denote the marginal pmf of X_1 and X_2, respectively.
- Let x_1 be a point in the support of X_1, $p_{X_1}(x_1) > 0$, we have

$$\mathbb{P}(X_2 = x_2 | X_1 = x_1) = \frac{\mathbb{P}(X_2 = x_2, X_1 = x_1)}{\mathbb{P}(X_1 = x_1)} = \frac{p_{X_1,X_2}(x_1, x_2)}{p_{X_1}(x_1)},$$

  for all x_2 in the support $\mathcal{S}_{X_2}$ of X_2.

Conditional Distributions — Discrete

- **Conditional pmf** of the discrete type of random variable X_2 given that the discrete type of random variable $X_1 = x_1$:

$$p_{X_2|X_1}(x_2|x_1) = \frac{p_{X_1,X_2}(x_1, x_2)}{p_{X_1}(x_1)},\ x_2 \in \mathcal{S}_{X_2}.$$

- Similarly,

$$p_{X_1|X_2}(x_1|x_2) = \frac{p_{X_1,X_2}(x_1, x_2)}{p_{X_2}(x_2)},\ x_1 \in \mathcal{S}_{X_1}.$$

- $p_{1|2}(x_1|x_2)$, $p_{2|1}(x_2|x_1)$, $p_1(x_1)$ and $p_2(x_2)$.

Conditional Distributions — Continuous

- Let X_1 and X_2 have the joint pdf $f_{X_1,X_2}(x_1, x_2)$ and the marginal pdf $f_{X_1}(x_1)$ and $f_{X_2}(x_2)$, respectively.
- **Conditional pdf** of the continuous type of random variable X_2 given that the continuous type of random variable $X_1 = x_1$:

$$f_{X_2|X_1}(x_2|x_1) = \frac{f_{X_1,X_2}(x_1, x_2)}{f_{X_1}(x_1)},$$

when $f_{X_1}(x_1) > 0$.

Conditional Distributions — Continuous

- For any fixed x_1, $f_{X_2|X_1}(x_2|x_1)$ is nonnegative and that

$$\int_{-\infty}^{\infty} f_{X_2|X_1}(x_2|x_1)dx_2 = \int_{-\infty}^{\infty} \frac{f_{X_1,X_2}(x_1,x_2)}{f_{X_1}(x_1)}dx_2$$
$$= \frac{1}{f_{X_1}(x_1)} \int_{-\infty}^{\infty} f_{X_1,X_2}(x_1,x_2)dx_2$$
$$= \frac{1}{f_{X_1}(x_1)} f_{X_1}(x_1) = 1.$$

- Since each of $f_{X_2|X_1}(x_2|x_1)$ and $f_{X_1|X_2}(x_1|x_2)$ is a pdf of one random variable, each has all the properties of such a pdf.

Conditional Distributions

- If the random variables are continuous, the conditional probability that $a < X_2 < b$ given that $X_1 = x_1$,

$$\mathbb{P}(a < X_2 < b | X_1 = x_1) = \int_a^b f_{2|1}(x_2|x_1) dx_2.$$

- Similarly, the conditional probability that $c < X_1 < d$ given that $X_2 = x_2$,

$$\mathbb{P}(c < X_1 < d | X_2 = x_2) = \int_c^d f_{1|2}(x_1|x_2) dx_1.$$

Conditional Distributions

- If $u(X_2)$ is a function of X_2, the conditional expectation of $u(X_2)$ given that $X_1 = x_1$ is given by

$$\mathbb{E}[u(X_2)|x_1] = \int_{-\infty}^{\infty} u(x_2) f_{2|1}(x_2|x_1) dx_2.$$

- $\mathbb{E}(X_2|x_1)$ is the mean and $\mathbb{E}\{[X_2 - E(X_2|x_1)]^2|x_1\}$ is the variance of the conditional distribution of X_2 given $X_1 = x_1$.
- Of course, we have

$$Var(X_2|x_1) = \mathbb{E}(X_2^2|x_1) - [\mathbb{E}(X_2|x_1)]^2.$$

An Example

- Let X_1 and X_2 have the joint pdf

$$f(x_1, x_2) = \begin{cases} 6x_2 & 0 < x_2 < x_1 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

  Find $\mathbb{E}(X_2|x_1)$.
  The marginal pdf of X_1 is $f_1(x_1) = \int_0^{x_1} 6x_2 \, dx_2 = 3x_1^2$, $0 < x_1 < 1$. The conditional pdf of X_2 given $X_1 = x_1$ is

$$f_{2|1}(x_2|x_1) = \frac{6x_2}{3x_1^2} = \frac{2x_2}{x_1^2}, \quad 0 < x_2 < x_1.$$

  The conditional mean is $\mathbb{E}(X_2|x_1) = \int_0^{x_1} x_2 \frac{2x_2}{x_1^2} dx_2 = \frac{2}{3}x_1$, $0 < x_1 < 1$. Now $\mathbb{E}(X_2|X_1) = 2X_1/3$ is a random variable.
- **Important!** $\mathbb{E}(X_2|x_1)$ is a function of x_1, then $\mathbb{E}(X_2|X_1)$ is a random variable with its own distribution, mean and variance.

A Theorem

- Theorem 2.3.1: (a) $\mathbb{E}[\mathbb{E}(X_2|X_1)] = \mathbb{E}(X_2)$
  (b) $Var[\mathbb{E}(X_2|X_1)] \leq Var(X_2)$
- Both X_2 and $\mathbb{E}(X_2|X_1)$ have the same mean μ_2. We could use either of the two random variable to guess at the unknown μ_2. But we would put more reliance in $\mathbb{E}(X_2|X_1)$.